UNIT 3 · TOPIC 4(B)

Testing of Threshers

Syllabus topic: Testing of harvesting and threshing equipment, grain and straw combines

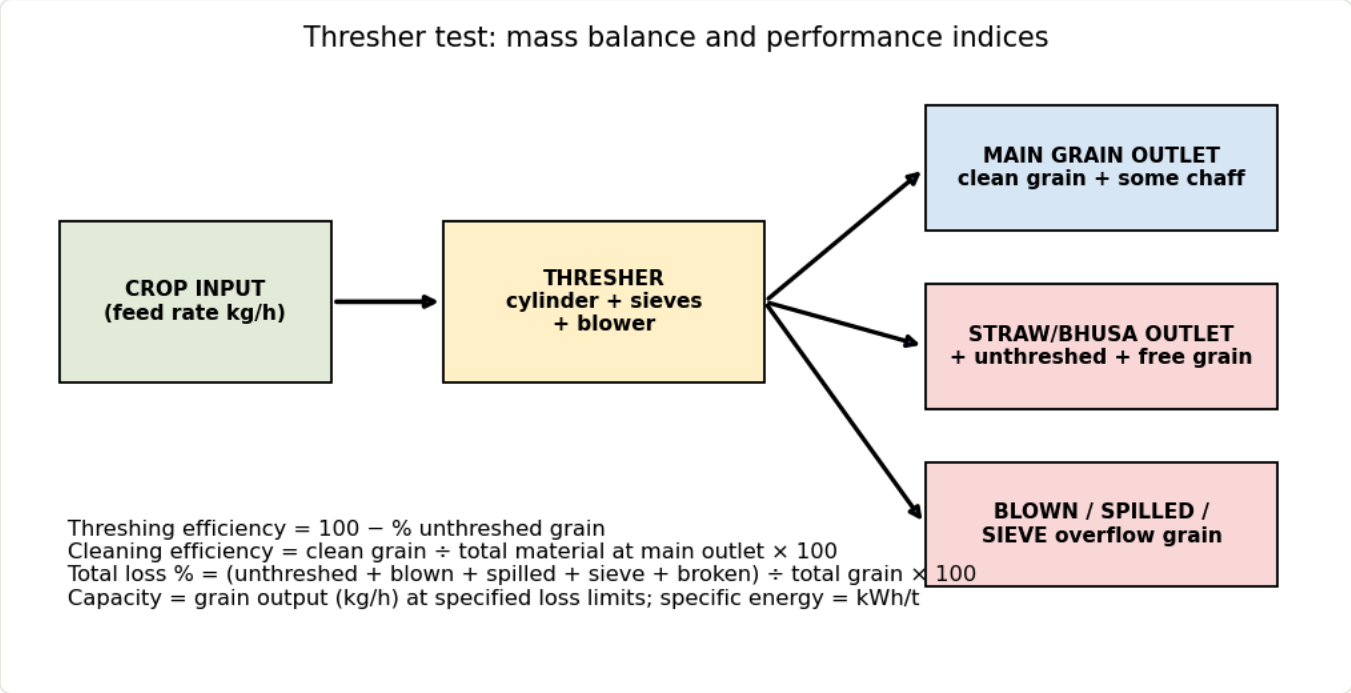
Indian standards for power threshers

Standard	Covers
IS 6284	Test code
IS 9020	Safety requirements
IS 9129	Feeding-chute safety (Unit 2, Topic 8; Unit 5)

BIS has withdrawn IS 9129 and folded it into IS 9020 (Power Threshers — Safety Requirements, revised 2002). Exam keys may cite either.

Test conditions

- Test at the recommended cylinder speed, with the crop at the specified moisture.
- Run **≥ 3 replications** at increasing feed rates.



Thresher test — mass balance and performance indices

What is recorded

Total grain input (grain from ALL outlets); unthreshed grain (in straw + main outlet + sieve overflow); clean grain and total material at the main outlet; blown/chaffed, spilled/scattered, sieve-overflow and visible broken grain; grain output (kg/h); power or energy used. Formulas and a worked example are in Topic 7a.

Practice MCQs

Q86. The Indian test code for power threshers is:

(a) IS 6284 (b) IS 6316
(c) IS 4468 (d) IS 730

Q87. Safety requirements for power threshers are covered in:

(a) IS 9020 (b) IS 2062
(c) IS 1239 (d) IS 808

Answers

Q86 — (a) IS 6284
Q87 — (a) IS 9020

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.